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STRAIN SENSING DEVICE AND ELECTRIC ACTUATOR HAVING THE SAME

Abstract

A strain sensing device includes a load body, a pair of strain gauges, and a cable connector. The load body has a chamber, an opening, and a mounting cover. The chamber is communicated to outside of the load body through the opening. A pair of detection surfaces are arranged on opposite sides of the chamber. The opening and the mounting cover are located on opposite sides of the chamber. The cable connector includes a connector head arranged on the mounting cover, and a cable threading the connector head and electrically connected to the strain gauges. When the load body is affected by an external force, the strain gauges respectively detect a deformation of the detection surfaces. The strain sensing device is therefore using alone as a load cell or in conjunction with an actuator and easily installing, disassembling and connecting an external wire.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to electric actuator, particularly to an electric actuator with an externally wired strain sensing device.

Description of Related Art

[0002] Electric actuators are widely used in industrial, medical, household beds, massage chairs, fitness equipment, rehabilitation equipment, door openers, window openers, lifting mechanisms and other devices and function to push objects weighting tens of kilograms or even thousands of kilograms to achieve lifting or changing angle. Most of the existing electric actuators use an electric motor coupled with a transmission mechanism and a reduction mechanism to drive the lead screw to rotate, and drives the screw-connected telescopic tube to linearly extend or retract by the lead screw.

[0003] However, electric actuators have their own load limits, most electric actuators cannot measure the weight of a load. Especially the weights carried by industrial or agricultural electric actuators are so heavy that may exceed the upper limit of the electric actuators to easily cause deformation and damage of the telescopic rod or other mechanisms. To avoid this situation, strain gauges are currently installed on electric actuators to measure the weight of the load. The signal line of the strain gauge is arranged inside the electric actuator, which causes inconvenience in installing and disassembling the strain gauge. Besides, users cannot replace the strain gauge according to their needs, or use software other than the one specified by the designer to cooperate with the inspection.

[0004] In view of the above, the inventor seeks to overcome the aforementioned drawbacks associated with the current technology and aims to provide an effective solution through extensive researches along with utilization of academic principles and knowledge.

SUMMARY

[0005] The primary objective of the present disclosure is to provide an externally wired strain sensing device, which is used as a load cell, or used with an electric actuator, easily installed or disassembled, and easily connected to an external wire.

[0006] To accomplish the aforementioned objective, the present disclosure provides a strain sensing device having a load body, a pair of strain gauges and a cable connector, the load body defines a chamber and an opening and has a mounting cover, the chamber communicates to outside of the load body through the opening, a pair of detection surfaces is arranged on opposite sides of the chamber, the opening and the mounting cover is located on opposite sides of the chamber, each of the strain gauges is arranged on each of the detection surfaces, the cable connector has a connector head arranged on the mounting cover and a cable threading the connector head and electrically connected to the strain gauges, when the load body is affected by an external force, each of the strain gauges is configured to detect a deformation of each of the detection surfaces.

[0007] Another aspect of the present disclosure provides that the load body defines a notch, the notch is formed on part of a periphery of the mounting cover and located on same side of the chamber with the mounting cover.

[0008] Another aspect of the present disclosure provides that the notch is arc-shaped or inverted U-shaped.

[0009] Another aspect of the present disclosure provides that further has a sealing member, the sealing member seals the opening and wraps the strain gauges.

[0010] Another aspect of the present disclosure provides that further has a circuit board, the circuit board is arranged in the chamber and located between the strain gauges, the circuit board is electrically connected to the strain gauges and the cable.

[0011] To accomplish the aforementioned objective, the present disclosure further provides an electric actuator having a gear box, a motor, a leadscrew, a telescopic tube, and the strain sensing device as mentioned above, the motor is connected to the gear box, part of the leadscrew is accommodated in the gear box and driven by the motor, the other part of the leadscrew extends outside of the gear box, the telescopic tube is threadedly connected to the leadscrew for transmission, the strain sensing device is configured to be detachably connected to the gear box such that an end of the leadscrew is accommodated in the load body, when the telescopic tube is driven to apply a thrust force such that an acting force of the thrust force transmits to impact the load body, each of the strain gauges is configured to detect a deformation of each of the detection surfaces.

[0012] Another aspect of the present disclosure provides that the load body defines a pit and has a stopping ring, the pit is recessed from an end surface of the load body, the stopping ring extends outward from an outer edge of the load body.

[0013] Another aspect of the present disclosure provides that the gear box has a cover shell, the cover shell has a step and defines a through hole, the step is formed on a periphery of the through hole, the load body penetrates the through hole, the stopping ring abuts against the step, the end of the leadscrew accommodated in the pit.

[0014] Another aspect of the present disclosure provides that further has an end cap, the end cap is sleeved and fixed on the leadscrew, the stopping ring is clamped and fixed between the end cap and the step.

[0015] Another aspect of the present disclosure provides that the end cap has a positioning block, the load body has an embedded groove, the positioning block is embedded in the embedded groove.

[0016] In the strain sensing device of the present disclosure, the strain gauges and the circuit boards are easily assembled through the opening communicated to the chamber. Through the Wheatstone bridge formed by the strain gauges and the circuit boards is integrated with the mating connector, and the mounting cover is arranged on the side opposite to the opening of the chamber for cable connector installing, the strain sensing device of the present disclosure can lead signal wires to outside through the cable connector for use according to different needs and the strain sensing device can be easily assembled or disassembled. The strain sensing device is therefore can use alone as the load cell or use in conjunction with an actuator such as electric actuator. That is each of the strain gauges can detect the changes in resistance value caused by the slight deformation of each of the detection surfaces when the load body is affected by the external force, the changes in resistance value is converted into the deformation of each of the detection surfaces through the Wheatstone bridge, and then the value of the deformation is transmitted to the external through the cable connector for the user to read or for the controller to receive.

[0017] The electric actuator of the present disclosure installs the strain sensing device corresponding to the leadscrew on the gear box or the end of the inner tube, so that when the motor drives the leadscrew to rotate by the transmission mechanism to drive the inner tube to extend from the outer tube, the strain gauges can detect the changes in resistance value caused by the slight deformation of each of the detection surfaces, the changes in resistance value is converted into the deformation of each of the detection surfaces through the Wheatstone bridge to measure the

loading weight of the electric actuator, and the user can quickly change the position of the strain sensing device according to different needs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a perspective appearance view of the load body of the present disclosure;

[0019] FIG. 2 is an exploded view of the strain sensing device of the present disclosure;

[0020] FIG. 3 is a perspective appearance view of the strain sensing device of the present disclosure;

[0021] FIG. 4 is a cross-sectional side view of the strain sensing device of the present disclosure;

[0022] FIG. 5 is a perspective appearance view of the electric actuator in first embodiment of the present disclosure;

[0023] FIG. 6 is a partially exploded view of the electric actuator in first embodiment of the present disclosure; and

[0024] FIG. 7 is an exploded view of the end cap and the load body in first embodiment of the present disclosure; and

[0025] FIG. 8 is a partially cross-sectional view of the electric actuator in first embodiment of the present disclosure; and

[0026] FIG. 9 is a top view of the electric actuator in second embodiment of the present disclosure.

DETAILED DESCRIPTION

[0027] It is to be understood that the terms for indicating positions and the location relation, for example “front”, “rear”, “left”, “right”, “front end”, “rear end”, “distal end”, “longitudinal direction”, “lateral direction”, “vertical direction”, “top” and “bottom”, are based on the positions and the location relation disclosed in the drawings, and only used for disclosing the present disclosure and not used for indicating or implying the specified location of the device or the components or the specified structure and operation in certain location, thus the present disclosure is not intended to be limiting.

[0028] For example, the terms of “first”, “second”, “third”, “forth” and “fifth” are used for illustrating each unit, component, area, layer and/or part. The component, the unit, the area, the layer and/or the part are not limited by the terms. These terms are only used for separating the element, the assembly, the area, the layer, or the part. Unless being clearly indicated according to the whole specification, the terms for example “the first”, “the second”, “the third”, “the fourth” and “the fifth” are not used for implying the order or sequence.

[0029] The technical contents of the present disclosure will become apparent with the detailed description of embodiments and the accompanied drawings as follows. However, it shall be noted that the accompanied drawings are for illustrative purposes only such that they shall not be used to restrict the scope of the present disclosure.

[0030] The present disclosure provides a strain sensing device **100**. As shown in FIGS. 1, 2, 3, and 4, the strain sensing device **100** of the present disclosure includes a load body **10**, a pair of strain gauges **20** and a cable connector **30**.

[0031] The load body **10** is made of metal alloy such as stainless steel or aluminum alloy to have good mechanical strength, but the present disclosure does not specifically limit what the material of the load body **10**. In the embodiment, the load body **10** extends along a longitudinal direction thereof and is generally cylindrical in shape, but the present disclosure is not limited to this specific shape. The shape of the load body **10** can be changed and adjusted according to needs. The load body **10** has a chamber **11**, an opening **12**, a pair of detection surfaces **111**, and a mounting cover **13**. The chamber **11** is communicated to an outside space of the load body **10** through the opening **12**. In detail, the chamber **11** is a blind hole. In the embodiment, the chamber **11** has four inner

surfaces mutually perpendicular to each other. The two opposite inner surfaces (depicted as a top surface and a bottom surface of the chamber **11** in FIG. **1**) are respectively the detection surface **111**, and the other two opposite inner surfaces are respectively a configuration surface **112**. In detail, the detection surfaces **111** are configured to arranged relatively parallel and opposite to each other within the chamber **11**. The opening **12** is located on one side of the chamber **11**, the mounting cover **13** is located on other side of the chamber **11** opposite to the opening **12**.

[0032] The strain gauge **20** includes an insulating sheet (not labeled in figures) and a metallic sensitive fence (not labeled in figures). The strain gauge **20** is used to measure a strain of an object deformed by an external force, while the metallic sensitive fence also deforms accordingly to cause corresponding changes in resistance value of the object. Each of the strain gauges **20** is arranged on each of the detection surfaces **111** inside the chamber **11** through the opening **12** to measure a deformation of each of the detection surfaces **111**. In detail, the strain sensing device **100** of the present disclosure further includes a pair of circuit boards **40**. Each of the circuit boards **40** is arranged on each of the configuration surfaces **112** inside the chamber **11** through the opening **12** and located between the strain gauges **20**. That is, each of the strain gauges **20** and each of the circuit boards **40** are arranged perpendicularly and staggered to each other. One of the circuit boards **40** is electrically connected in series with one of the strain gauges **20** to form a first half bridge, and the other circuit board **40** is electrically connected in series with the other strain gauge **20** to form a second half bridge. The first half bridge and the second half bridge are electrically connected in parallel to form a Wheatstone bridge. The Wheatstone bridge can measure the changes in resistance value of the object deformed by the external force to convert to a strain value of the object.

[0033] The cable connector **30** includes a connector head **31** and a cable **32**. The connector head **31** is detachably arranged on the mounting cover **13**. In the embodiment, the connector head **31** is screwed and fixed with an external thread on an internal thread of the mounting cover **13**, but the present disclosure is not limited to this. For example, the connector head **31** may also arranged on the mounting cover **13** by fastening, locking, buckling or bonding. The cable **32** is threading the connector head **31** and electrically connected to the strain gauges **20** and the circuit boards **40**. In detail, the cable **32** penetrates from one end of the connector head **31** and emerges from the other end of the connector head **31** to be located in the chamber **11** of the load body **10** and to form an electrical connection with the Wheatstone bridge. In the embodiment, although it is not shown in figures, the person having ordinary skill in the art (PHOSITA) can understand that the Wheatstone bridge connecting by the first half bridge and the second half bridge in parallel may electrically connect to a mating connector first, then the cable connector **30** can directly connect to the mating connector to achieve electrical connection when the cable connector **30** installed on the mounting cover **13**.

[0034] Therefore, when the load body **10** is affected by an external force, each of the strain gauges **20** can detect the changes in resistance value caused by the slight deformation of each of the detection surfaces **111**, and then convert the changes in resistance value into the deformation of each of the detection surfaces **111** through the Wheatstone bridge, and then transmit the value of the deformation to an external display (not shown in figures) or a controller (not shown in figures) through the cable connector **30** for an user to read or for the controller to receive. In addition, since the strain gauges **20** and the circuit boards **40** can be arranged inside the chamber **11** through the opening **12** for easy installing, and the Wheatstone bridge formed by the strain gauges **20** and the circuit boards **40** is integrated with the mating connector, the strain sensing device **100** of the present disclosure can lead signal wires to outside through the cable connector **30** for use according to different needs and the strain sensing device **100** can be easily assembled or disassembled. The strain sensing device **100** is therefore can use alone as a load cell or use in conjunction with an actuator such as electric actuator to monitor the strain value.

[0035] Details are provided as follows. Please refer FIG. **2**. The load body **10** defines a notch **14**

communicated to the chamber **11**. In detail, the notch **14** is formed on part of a periphery of the mounting cover **13** and located on same side of the chamber **11** with the mounting cover **13**. In the embodiment, the notch **14** is inverted U-shaped, but the present disclosure is not limited to this specific shape. For example, the notch **14** may also be arc-shaped, inverted V-shaped or other shapes. It is worth noting that an inverted U-shaped opening of the notch **14** faces to one of longitudinal end of the load body **10** in the embodiment. Thus, since the notch **14** is communicated to the chamber **11** and arranged symmetrically with respect to the opening **12**, and the inverted U-shaped opening of the notch **14** faces to one of longitudinal end of the load body **10**, the opening **12** and the notch **14** can provide a space allowing the load body **10** to deform when the load body **10** is affected by the external force. The load body **10** is therefore prevented from uneven deformation such that it effectively improves a sensing effect of the strain gauges **20**.

[0036] Please refer FIGS. **1**, **2**, **3**, and **4**. The load body defines a penetrated hole **15**. The penetrated hole **15** runs through the load body **10** and is used for the insertion of a fixing equipment, a pin, or a screw mating to the strain sensing device **100**. In the embodiment, the strain gauges **20** are respectively arranged along a direction perpendicular to an extended direction that of the penetrated hole **15**, but the present disclosure is not limited to this direction. For example, the strain gauges **20** may also be respectively arranged along a direction parallel to an extended direction of the penetrated hole **15**.

[0037] Please refer FIG. **3** and FIG. **4**. The strain sensing device **100** of the present disclosure further includes a sealing member **50**. The sealing member **50** can be an elastomer such as silicone or rubber. In the embodiment, the sealing member **50**, for example, is silicone SE9176, but the present disclosure is not limit to this silicone. The sealing member **50** seals the opening **12**, and wraps the mating connector and the Wheatstone bridge formed by the strain gauges **20** and the circuit boards **40**. In other embodiment, the sealing member **50** may also seal the notch **14** to completely seal the chamber **11** and block the communication of the chamber **11** with the outside space. Therefore, the sealing member **50** can effectively protect each member in the chamber **11**, and effectively achieve high air-permeability, insulation, heat resistance, and waterproof inside the strain sensing device **100** at the same time.

[0038] The present disclosure further provides an electric actuator. As shown in FIGS. **5**, **6**, **7**, and **8**, those figures show the first embodiment of the electric actuator of the present disclosure. The electric actuator of the present disclosure includes a gear box **600**, a motor **700**, a leadscrew **800**, a telescopic tube **900**, and the strain sensing device **100** as mentioned above.

[0039] The gear box **600** includes a base seat **610**, a cover shell **620**, and a transmission mechanism **630**. The cover shell **620** provides cover and is fixed relative to the base seat **610**. The transmission mechanism **630** is accommodated in the base seat **610** and the cover shell **620**.

[0040] The motor **700** is connected to the gear box **600**. In detail, the motor **700** is arranged on a side of the base seat **610** away from the cover shell **620**. The motor **700** has a driving shaft **710**. The driving shaft **710** is power-transmitted connected to the transmission mechanism **630**.

[0041] A part of the leadscrew **800** is accommodated in the gear box **600** and driven by the motor **700**, the other part of the leadscrew **800** extends outside of the gear box **600**. In detail, the leadscrew **800** is arranged on the side of the base seat **610** away from the cover shell **620** and parallel adjacent to the motor **700**. The leadscrew **800** is power-transmitted connected to the transmission mechanism **630** such that the motor **700** can drive the leadscrew **800** to rotate through the transmission mechanism **630**.

[0042] The telescopic tube **900** is threadedly connected to the leadscrew **800** for transmission. In detail, the telescopic tube **900** includes an outer tube **910**, an inner tube **920**, and a supporting seat **930**. The inner tube **920** is sleeved on a periphery of the leadscrew **800** and threadedly connected to the leadscrew **800**, the outer tube **910** is sleeved along the periphery of the leadscrew **800** for the inner tube **920** threading inside, so that the leadscrew **800** can drive the inner tube **920** to linearly move along an axial direction relative to the outer tube **910** when rotated. The supporting seat **930**

is fixed, by screw connection, on an end of the inner tube **920** away from the gear box **600**. The supporting seat **930** defines a penetrated hole **931** used for receiving the insertion of a fixing equipment, a pin, or a screw mating to the electric actuator.

[0043] The specific members and structures of the strain sensing device **100** are the same as mentioned above, so the description thereof is omitted here for brevity. In the embodiment, the strain sensing device **100** is configured to be detachably connected to the gear box **600** such that an end of the leadscrew **800** is accommodated in the load body **10**. In detail, the load body **10** defines a pit **16** and has a stopping ring **17**, and the cover shell **620** has a step **621** and defines a through hole **622**. Please refer FIG. 2 and FIG. 4, the pit **16** is recessed from an end surface of the load body **10**, the stopping ring **17** extends outward from an outer edge of the load body **10** and located on the end surface formed on the load body **10**. Please refer FIG. 6 and FIG. 8, the through hole **622** is arranged on the cover shell **620** corresponding to the end of the leadscrew **800**, and the step **621** is formed on a periphery of the through hole **622**. Please refer FIG. 8, when the electric actuator is completely assembled, the load body **10** penetrates the through hole **622** of the cover shell **620** such that the stopping ring **17** of the load body **10** abuts against the step **621** of the cover shell **620** to accommodate the end of the leadscrew **800** in the pit **16** of the load body **10**.

[0044] Therefore, when the telescopic tube **900** is driven to apply to a thrust force, that is the motor **700** drives the leadscrew **800** to rotate through the transmission mechanism **630** to drive the inner tube **920** to extend relative to the outer tube **910** such that an acting force of the thrust force transmits to impact the load body **10**, each of the strain gauges **20** of the strain sensing device **100** can detect the deformation of each of the detection surfaces **111** to measure a loading weight of the electric actuator.

[0045] Please refer FIGS. 6, 7, and 8, a thread portion **18** is formed on part of the outer edge of the load body **10**. Thus, through screwing a nut A on the thread portion **18** to press against an outside of the cover shell **620** by an end surface of the nut A, the load body **10** is therefore fixed on the cover shell **620**, but the present disclosure is not limit to this fixed way. For example, the load body **10** may also be fixed on the cover shell **620** by fastening, locking, buckling or bonding.

[0046] Details are provided as follows. The electric actuator of the present disclosure further includes an end cap **631**, a bushing **632**, and a bearing **633**. The end cap **631** includes a hollow plate **6311** and a hollow convex-ring **6312**. The end cap **631** is sleeved and fixed on the leadscrew **800**, and the stopping ring **17** is clamped and fixed between the end cap **631** and the step **621**. In detail, the bearing **633** is sleeved on the leadscrew **800** and supported between the end cap **631** and the bushing **632**, and the hollow plate **6311** and the hollow convex-ring **6312** are jointly sleeved on the bearing **633**. Thus, the stopping ring **17** is clamped and fixed between the hollow plate **6311** of the end cap **631** and the step **621** of the cover shell **620** along the longitudinal direction, and clamped and fixed between the hollow convex-ring **6312** of the end cap **631** and the cover shell **620** along a vertical direction (depicted as an up-and-down direction in FIG. 8). Therefore, when the electric actuator is mounted on a load, the acting force exerted to the leadscrew **800** will be transmitted to the load body **10** along the bushing **632**, the bearing **633**, the end cap **631** and the step **621** sequentially, so that the strain gauges **20** detect the changes in resistance value caused by the deformation of the load body **10**, and then the changes in resistance value is converted into the deformation of each of the detection surfaces **111** through the Wheatstone bridge, and the value of the deformation is transmitted to outside of the load body **10** through the cable connector **30**.

[0047] In addition, please refer FIG. 7, the end cap **631** includes a positioning block **6313**, the load body defines an embedded groove **19**. The positioning block **6313** is embedded in the embedded groove **19** to avoid the end cap **631** from rotating relative to the load body **10**. In detail, the positioning block **6313** extends from a side surface of the hollow plate **6311** facing to the hollow convex-ring **6312**, and the embedded groove **19** is formed on the end surface of the load body **10** forming the pit **16**. Thus, the load body **10** is positioned on the end cap **631** and unable to rotate when the positioning block **6313** is embedded in the embedded groove **19**, so that it is effective to

ensure a sensing accuracy and stability of the strain sensing device **100**. In the embodiment, a number of the positioning block **6313** is two, but the present disclosure is not limited thereto. For example, the number of positioning block **6313** may also be one or more than three.

[0048] It is noting that the strain sensing device **100** of the electric actuator of the present disclosure is not limit to the configuration of installing on the gear box **600**. Please refer FIG. 9, which is the second embodiment of the electric actuator of the present disclosure. The main difference between the second embodiment and the first embodiment mentioned above is that in the second embodiment the positions of the strain sensing device **100** and the supporting seat **930** are exchanged in comparison with that in the first embodiment. In detail, the strain sensing device **100** in the embodiment is screwed on the end of the inner tube **920** through the thread portion **18** of the outer edge of the load body **10**, and the supporting seat **930** is mounted on cover shell **620** of the gear box **600** to detect a strain change of the end of the inner tube **920** through the strain sensing device **100**. That is the user can change the position of the strain sensing device **100** according to different needs, the user can also choose to not dispose the strain sensing device **100** at either end of the inner tube **920** when not needs to detect strain changes.

[0049] In the strain sensing device **100** of the present disclosure, the strain gauges **20** and the circuit boards **40** are easily assembled through the opening **12** communicated to the chamber **11**. Through the Wheatstone bridge formed by the strain gauges **20** and the circuit boards **40** is integrated with the mating connector, and the mounting cover **13** is arranged on the side opposite to the opening **12** of the chamber **11** for cable connector **30** installing, the strain sensing device **100** of the present disclosure can lead signal wires to outside through the cable connector **30** for use according to different needs and the strain sensing device **100** can be easily assembled or disassembled. The strain sensing device **100** is therefore can use alone as the load cell or use in conjunction with an actuator such as electric actuator. That is each of the strain gauges **20** can detect the changes in resistance value caused by the slight deformation of each of the detection surfaces **111** when the load body **10** is affected by the external force, the changes in resistance value is converted into the deformation of each of the detection surfaces **111** through the Wheatstone bridge, and then the value of the deformation is transmitted to the external through the cable connector **30** for the user to read or for the controller to receive.

[0050] The electric actuator of the present disclosure installs the strain sensing device **100** corresponding to the leadscrew **800** on the gear box **600** or the end of the inner tube **920**, so that when the motor **700** drives the leadscrew **800** to rotate by the transmission mechanism **630** to drive the inner tube **920** to extend from the outer tube **910**, the strain gauges **20** can detect the changes in resistance value caused by the slight deformation of each of the detection surfaces **111**, the changes in resistance value is converted into the deformation of each of the detection surfaces **111** through the Wheatstone bridge to measure the loading weight of the electric actuator, and the user can quickly change the position of the strain sensing device **100** according to different needs.

[0051] It shall be understood that the present disclosure may have other types of embodiments, and a person with ordinary skills in the art of the technical field of the present disclosure may make various changes and modifications corresponding to the present disclosure without deviating the principle and substance of the present disclosure; however, such corresponding changes and modification shall be considered to be within the claimed scope of the present disclosure.

Claims

1. A strain sensing device, comprising: a load body, defining a chamber and an opening and comprising a mounting cover, the chamber communicated to outside of the load body through the opening, a pair of detection surfaces arranged on opposite sides of the chamber, the opening and the mounting cover located on opposite sides of the chamber; a pair of strain gauges, respectively arranged on each of the detection surfaces; and a cable connector, comprising a connector head

- arranged on the mounting cover and a cable threading the connector head and electrically connected to the strain gauges; wherein when the load body is affected by an external force, each of the strain gauges is configured to detect a deformation of each of the detection surfaces.
2. The strain sensing device according to claim 1, wherein the load body defines a notch, the notch is formed on part of a periphery of the mounting cover and located on same side of the chamber with the mounting cover.
3. The strain sensing device according to claim 2, wherein the notch is arc-shaped or inverted U-shaped.
4. The strain sensing device according to claim 1, further comprising a sealing member, the sealing member seals the opening and wraps the strain gauges.
5. The strain sensing device according to claim 1, further comprising a circuit board, the circuit board is arranged in the chamber and located between the strain gauges, the circuit board is electrically connected to the strain gauges and the cable.
6. An electric actuator, comprising: a gear box; a motor, connected to the gear box; a leadscrew, part of the leadscrew accommodated in the gear box and driven by the motor, the other part of the leadscrew extended outside of the gear box; a telescopic tube, threadedly connected to the leadscrew for transmission; and the strain sensing device according to one of claim 1, configured to be detachably connected to the gear box such that an end of the leadscrew is accommodated in the load body; Wherein when the telescopic tube is driven to apply a thrust force such that an acting force of the thrust force transmits to impact the load body, each of the strain gauges is configured to detect a deformation of each of the detection surfaces.
7. The electric actuator according to claim 6, wherein the load body defines a pit and comprises a stopping ring, the pit is recessed from an end surface of the load body, the stopping ring extends outward from an outer edge of the load body.
8. The electric actuator according to claim 7, wherein the gear box comprises a cover shell, the cover shell comprises a step and defines a through hole, the step is formed on a periphery of the through hole, the load body penetrates the through hole, the stopping ring abuts against the step, the end of the leadscrew accommodated in the pit.
9. The electric actuator according to claim 8, further comprising an end cap, the end cap is sleeved and fixed on the leadscrew, the stopping ring is clamped and fixed between the end cap and the step.
10. The electric actuator according to claim 9, wherein the end cap comprises a positioning block, the load body comprises an embedded groove, the positioning block is embedded in the embedded groove.
11. The electric actuator according to claim 6, wherein the load body defines a notch, the notch is formed on part of a periphery of the mounting cover and located on same side of the chamber with the mounting cover.
12. The electric actuator according to claim 11, wherein the notch is arc-shaped or inverted U-shaped.
13. The electric actuator according to claim 6, further comprising a sealing member, the sealing member seals the opening and wraps the strain gauges.
14. The electric actuator according to claim 6, further comprising a circuit board, the circuit board is arranged in the chamber and located between the strain gauges, the circuit board is electrically connected to the strain gauges and the cable.
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